

FlashIQ

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PROJECT PROBLEM STATEMENT

Discuss the problem or opportunity that is scheduled for resolution.

- Students struggle with studying and memorizing material quickly, especially when trying to cram before a test. Today, most flashcard studying apps aren't very comprehensive, or the high quality ones require you to pay a specific fee to get access to all their features. With this app, students or learners can have free, easy access to effective studying software that aids them in preparation.

CONCISE DESIGN OVERVIEW

After researching potential technologies you will identify the components of your solution. Develop a prototype or model of what your final solution would look like, providing abstract information about the individual components.

- Home screen - shows all of the user's made decks
- Quiz feature - user can test themselves using their flashcards and indicate whether they know or don't know the concept (correct v. wrong)
- Results screen - after user is done testing themselves, the result screen shows their accuracy percentage and score
- Manage Cards Screen - user can add new cards and remove existing ones. This allows the user to edit any set whenever they like.

SCOPE STATEMENT

SMART goals or objectives would include deliverables. Clarify as necessary what the scope includes and does not include.

- S - Create Java Swing app called FlashIQ to help people study more effectively
- M - All of the test cases pass without any difficulties. Then, the code will be put into a Github repository
- A - The whole app is one file, making it easy to download and complete
- R - The project should assess and show our Java skills we have learnt over the past 2 years - e.g. OOP, GUI, etc.
- T - The project will be completed on time, taking around 3-4 days.

TIMELINE

Draw a timeline, including milestones to serve as the basis for a work breakdown structure (WBS) and appropriate Gantt or PERT charts. For this class, you can refer to the Work Breakdown Excel Sheet. This can be a table with anticipated tasks listed for each school day of the project. You will submit a completed chart(or Record of Tasks) along with your final deliverables.

Day	Tasks Done
1	- Decide on an idea for the app - Set up the GitHub repository - Plan out how we want to code the app (e.g. the various screens)
2	- Create the basis for the project (e.g. a class for the Flashcards, methods to add and delete cards) - Create the home screen and quiz screen
3	- Implement the flashcard features (moving to different cards, marking them wrong or correct, tracking score) - Create the results screen - Make sure app passes test cases
4	- *If needed, polish up the app's appearance - Push source code to Github repo

TEST PLAN

Describe how you will test your solution. The plan should include testing the whole solution and testing individual components of the solution. Include details such as if you will automate testing or you will manually test with positive and negative test cases.

Test Case	Expected Result	Pass Criteria
Add and delete flashcards	The changes the user makes should reflect in the set. So, if they add a card to a set, it should show and the same goes for deletion.	If the user's actions are reflected in the flashcard set, then this test is passed.

Full quiz session	The user should be able to take a full quiz using their flashcard set. They should be able to shuffle the cards, flip them to reveal answers, mark them right or wrong, and be able to see the score at the end of the assessment.	If the user can successfully do all of these things, then this test is passed.
Empty deck	The user should be able to remove all cards from a deck, and then try to start a quiz with it. The app should not crash, but the quiz should not start.	If the user can do this, this test is also passed.

RISKS AND CONTINGENCY PLANS

List known and potential risks by estimated probability, with mitigation plans.

- Data gets lost when the app is closed, high, try to make sure all data is saved in some sort of folder
- Variance in screen sizes which makes app look weird, medium, use setSize or setPreferredSize to make sure app appearance is nice and organized
- UI doesn't look good, medium, if we run out of time we will try our best to make the UI look as nice as possible. The main priority is making sure the code works.